

1. Write a program to solve the questions using Forward Chaining.

a. Knowledge Base:

$P \rightarrow Q$

$L \wedge M \rightarrow P$

$A \wedge B \rightarrow L$

A

B

M

Conclusion:

Q

b. Knowledge Base:

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow D$

A

E

$D \wedge E \rightarrow F$

Conclusion: F

2. Write a program to solve the questions using Backward Chaining.

a. Knowledge Base:

$P \rightarrow Q$

$R \rightarrow Q$

$A \rightarrow P$

$B \rightarrow R$

A

B

Conclusion:

Q

b. Knowledge Base:

$A \rightarrow B$

$B \wedge C \rightarrow D$

$E \rightarrow C$

A

E

Conclusion:

D

3. Write a program to solve the questions using the method Resolution and proof of refutation/contradiction.

a. Knowledge Base:

$P \vee Q$

$P \rightarrow R$

$Q \rightarrow S$

$R \rightarrow S$

Conclusion

S

b. Knowledge Base:

$P \rightarrow Q$

$Q \rightarrow R$

$S \rightarrow \neg R$

P

Conclusion:

S